

design of light emitting diode (LED) boards and systems, as per the specifications and architecture given. They work on the optimum design solution and make sure it meets specifications within boundary conditions of cost and timeline. They are also responsible for preparing test samples, test plans, design documents and to maintain test set-up in the development. On a regular basis, they interact with quality teams for validating platform, module or product designs.

This function, as a designer or design assistant, requires closely working with component suppliers for the right selection and electronic manufacturing service (EMS) houses for engineering/pilot runs and test labs to get the required certifications. A designer's support in project estimation, planning and risk management activities is highly valuable. Overall, this position needs high level of domain knowledge and networking ability in a cross-functional environment.

On the other hand, a design technician deals with printed circuit board (PCB) mounting, sample making, and test and documentation as per requirements with respect to LED lamps.

Skills. The most required skill for a designer in the lighting industry is proficiency in PCB computer-aided design (CAD) software, such as those by Altium or Proteus. Along with good analytical skills, a designer is also expected to understand technical specifications of LEDs and PCBs. Basic understanding of the EMS process, its critical process parameters and ability to deal with external suppliers, particularly PCB/EMS, are some other skills that, if developed, can keep you ahead in the competition and possibly reward you with a great job. Good communication skills are also important for meetings and presentations.

A design technician is expected to have strong fundamentals in elec-

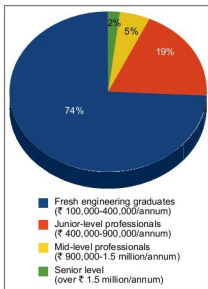


Fig. 1: Salary break-up for lighting professionals (Data courtesy: TimesJobs.com)

tronics as well as good communication skills.

Educational qualification. Bachelor of Engineering (B.E.) or Bachelor of Technology (B.Tech.) in electronics engineering/electronics and communication engineering is required.

You could be a marketing and sales officer

Lighting sales representatives are at the helm for selling lighting products. Some work as employees of lighting manufacturing organisations, but most are employed by independent agencies that have contracts to sell products of several lighting companies in a specific city, state, region or country. This area is where majority of the jobs in lighting sector are available. An engineering degree is a pre-requisite for most companies.

Roles and responsibilities. Positions in this side of lighting domain are usually titled sales engineer, lighting application specialist, sales officer, lighting sales representative, sales executive and lighting sales engineer, among others. Generally, a sales engineer is expected to provide lighting design and consultancy support in lighting projects. He or

she must co-ordinate and work with the sales team at the front-end, and the marketing and technical teams at the back-end to bring the best lighting solutions to customers, faster than competitors. An engineer is expected to achieve growth in sales through effective networking within assigned territory. The area of coverage is usually allocated after posting, and this job requires extensive travelling, at least 50 per cent of the time. He or she must also have good connections and contacts with architects to promote and introduce new products.

Skills. An engineer should be conversant with and have aptitude for lighting design and application as well as lighting software. With a good grip on communication, he or she should be able to interact with architects, engineering consultants and professional end-users. Basic lighting design skills, good business development skills, ability to support regional sales teams in lighting projects under supervision, knowledge of AutoCAD, good interpersonal skills and analytical skills (to keep track of competition activity in the market) are some other aspects that recruiters look for. Experience of working with architects and consultants is an added advantage. Recruiters prefer engineers with good interpersonal skills and skillful organisation of tasks, along with the ability to work with little or no supervision.

Educational qualification. B.E. or B. Tech. in electrical or electronics is required. An MBA is also preferred by some companies.

You could also be a test engineer

Companies who develop and manufacture lighting fixtures and electronics for domains such as automotive industry usually hire test engineers, although these jobs are limited.

Roles, responsibilities and skills expected. A test engineer will be in charge of LED printed circuit board